Manual

Monitoring of Machine Data (MOC)

MDE-MMD 8.1

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Contents

1 Overview of Machine Data Monitoring

Possible fields of application

The function package “Machine Data Monitoring” allows for the data recorded in the system to be evaluated with respect to the current status of machines and workplaces as well as in relation to the posted status durations.

Implementation notes

You use the function package if you wish:

- to monitor your machines and workplaces
- to evaluate the recorded times and durations in relation to machines/workplaces
- to evaluate the hours occurred in a special sector/department over a specific period of time in relation to different statuses

Integration

The machine data monitoring evaluates machines and workplaces. The times recorded in the machine data collection module are generally used for this purpose.

Functions

- Workplace/machine overview
    - Tabular workplace/machine overview to represent the current machine status. Display of a picture of the machine. Detailed views, e.g. list of registered staff, operations, tools or resources (available if relevant functions from these HYDRA areas are used at the same time as MDE (machine data collection)). Comparison of actual and target quantities relating to orders, target and actual cycles as well as target and actual stroke numbers.
- Cycle progression
    - Tabular and graphic evaluation on the actual cycle of a machine over configurable periods; can be displayed in "seconds per piece" or "pieces per minute“.
- Machine time profile
    - Graphic machine time profile with real-time presentation of the downtime/production performance of several machines over configurable periods including zoom function.
- Downtime profile
    - Drill-down function within the machine time profile for detailed presentation of the downtime performance of a machine
- Machine history

- Machine history for the tabular presentation of machine-related and order-related postings performed at a machine/workplace.

2 Workplaces/Machines

1.1 Summary

Menu	Production Facility Management → Current Information → Workplaces/Machines
Transaction code	wpov
Function authorization	wpov

Utilization

The workplace overview function is a report for production management. It aims at users from production scheduling and monitoring, schedulers, foremen, operators or all MOC users who would like to get a comprehensive overview of the production situation at certain workplaces/machines or a complete organizational unit.

Integration

At the push of a button, the workplace overview provides all pieces of information relevant for workplaces. In addition to master data, the function also provides the data required to control the production process, such as

- current workplace/machine status
- operations currently running at the workplace/machine
- currently used tools and resources
- cycle progression of the shift (for machines with clocked production)
- output per shift: quantities, durations

Selection criteria

The application provides the following selection criteria:

Workplace from... to ...

This selection criterion refers to the workplace within the machine/workplace master. Wildcards (*) can be used.

Group from ... to ...

This selection criterion refers to the group in the machine or workplace master. All workplaces/machines assigned to the selected group are displayed. Wildcards may be used.

Evaluation/report group

This selection criterion refers to the evaluation groups. All workplaces/machines assigned to the selected evaluation/report group are displayed.

Designation

This field refers to the designation of machines and workplaces within the machine master data. Only the machines matching the specified character string are displayed. Wildcards (*) can be used in this field.

Short name

This selection criterion refers to the short name of the machines within master data. All machines or workplaces matching the entered character string are displayed. It is possible to use wildcards.

Responsibility area

This selection criterion refers to the responsibility area within the workplace/machine master. Please respect that only machines are displayed, for which the user is authorized by the corresponding responsibility areas.

Company

This selection criterion refers to the company defined in the machine or workplace master. All workplaces/machines assigned to the selected company are displayed. Wildcards may be used.

Cost center

This selection criterion refers to the cost center defined in the machine or workplace master. All workplaces/machines assigned to the selected cost center are displayed. You can also run a search using wildcards.

Status

This selection criterion refers to the current machine or workplace status. All machines or workplaces, which are currently assigned to the selected status, are displayed.

Status longer than

This selection criterion refers to the current status of machines or workplaces. All machines or workplaces are shown that are currently assigned to the selected status and that are assigned to this status for a longer period than the one specified.

If several selection criteria are used overlapping results are displayed in the workplace overview.

"Workplace overview" detail application

The detail application “workplace overview” shows all workplaces subject to the selections made within the selection panel. The display is refreshed cyclically every three minutes. The current status, workplace information, shift quantities, cycles and cycle figures are presented. The following list describes the data available in the table. If this data is not displayed by default it can be added using the column selection function.

In addition operation data is also shown, provided that an operation is currently logged on. In case several operations are logged on to a machine, only the first operation is shown in the detail application.

Status

The "status" column summarizes the different statuses and presents them as an "LED". Coloring is as follows:

Light green	Status with RPA 11 (normally "Production")	
BLUE	Status with RPA 7 (normally "Setup")	
RED	Status 30000 (normally "Not assigned")	
GRAY	Status 20000 or status with RPA 12	(normally break/no shift
Yellow	Status < 10000 and RPA <> [7,11,12]	other statuses/downtimes

Master data

Workplace

Unique ID defined within the workplace configuration.

Short name

Machine designation as defined in the workplace configuration

Designation

Long text/comment on the machine as defined in the workplace configuration.

Groups

Group according to workplace configuration to which the machine has been assigned.

Cost centers

Cost center as defined in the workplace configuration

Company

Company as defined in the workplace configuration.

Responsibility area

Responsibility area required to view this workplace as it is defined in the workplace configuration

Model

Workplace model according to workplace configuration.

Type

Workplace type according to the workplace configuration.

Status

Status

Status number of the status that is currently active at the workplace. Color of the currently active status according to configuration.

Status designation

Status designation of the status that is currently active at the workplace.

Status since

Date when the status was assigned.

Status since

Point in time when the status was assigned.

Duration so far

Present duration of the status that is currently available at this workplace.

Expected total duration

Expected status duration recorded by the employee when assigning the status at the terminal or that is defined in the status configuration.

Expected end

“Status since” + expected duration; synchronized with the Gregorian calendar.

Expected remaining runtime

“Expected end“ minus “now“ Please note: if the remaining runtime is negative the expected end is already overdue. In this case, the field is highlighted in red.

Shift quantities primary quantity unit/secondary quantity unit/tertiary quantity unit/base quantity unit

Yield

Yield that has been posted so far at the selected workplace within the current shift.

Scrap

Scrap that has been posted so far at the selected workplace in the current shift.

Rework

Rework quantity that has been posted so far at the selected workplace in the current shift.

Open quantity

Open quantity that has been posted so far at the selected workplace within the current shift.

Unit

Unit of primary quantity

Cycle

Target cycle

Current target cycle at the workplace.

If an operation is logged on to the machine the target cycle defined for the operation is displayed in seconds per cycle. There is no target cycle for machines to which no OP is currently logged on. In this case, “0” is entered in the “target cycle” field.

Actual cycle

Current actual cycle of the workplace

Colored display of the actual cycle relating to the configured cycle parameters.

Difference (%)

The difference in% is calculated according to the following formula: (target cycle - actual cycle) / target cycle * 100%. If the actual cycle is slower than the target cycle, the difference is indicated in negative values, otherwise positive values are shown. See below for coloring.

Actual cycle (OP)

The actual cycle (OP) is a value referring to the order. The values used for the calculation all refer to order logons and, as a result, they are independent from the current machine status.

Formula: Actual cycle $_{OP}$ = RPA11 $_{OP}$/ (Yield $_{OP}$ / Partitioning $_{OP}$)

Difference (OP) (%)

The difference OP [%] column is computed according to the following formula: Difference$_{OP}$ = Abs((target cycle number – actual cycle number $_{OP}$) * 100) / target cycle number

Cycle number [1/min]

Target cycle number

1 / Target cycle

There is no target cycle for machines to which no OP is currently logged on. For this reason, the target stroke number is 0.

Actual cycle number

1 / actual cycle

Difference (%)

(target cycle number – actual cycle number) / target stroke number * 100%

Please note: For rounding reasons, the difference indicated here might deviate from the difference shown in the "cycle" category.

Actual cycle number (OP)

The actual cycle number (OP) is a value relating to orders. The values used for the calculation all refer to order logons and, as a result, they are independent from the current machine status. Formula: Actual cycle number $_{OP}$ = yield $_{OP}$/ (partitioning $_{OP}$ * RPA11 $_{OP}$)

Difference (OP) (%)

The difference OP column is computed by the following formula:

Difference$_{OP}$ = Abs((target cycle number – actual cycle number $_{OP}$) / target cycle number * 100)

Coloring of the "difference" column

Coloring for the "difference" column in the "cycle" category may be defined by master data (menu: master data > workplaces/machines > cycle parameters) per machine for the upper/lower action limits or upper/lower tolerance limits. The signed value of the difference column is used for coloring. The value in the difference column is displayed in red if the tolerance limits are exceeded; the value is displayed in blue if the action limits are exceeded. The data is not displayed in color if no cycle parameters are defined.

"Image" detail application

The picture in the “image” detail application shows the picture of a machine from the machine configuration. The image of the machine selected in the “workplaces” detail application is displayed.

The following image formats are supported: jpg, gif, png, tif, bmp, ico, emf, and wmf. The pictures have to be filed in a directory that may be accessed via the path ID “MOCWPIMG” within the path configuration.

Further detailed information about the configuration can be found here.

"Operations logged on" detail application

The detail application “operations logged on” shows all currently registered operations that are currently logged on to workplaces/machines, which are selected in the detail application “workplaces”. The following list describes the data available in the table. If this data is not displayed by default it can be added using the column selection function.

Workplace

Workplace

Workplace to which the operation is logged on.

Order

Order

Order number of the operation.

Sequence

Sequence number of the OP (provided that sequences are used).

OP

Operation number

Split

Split number of the operation (provided that the split function is used).

SOP

Sub-operation umber (reserved)

OP designation

Designation of the operation

Article

Article number produced by the operation; taken over from operation data

Logon

Date

Date when the operation was last logged on to this workplace

Time

Time when the operation was last logged on to this workplace

Primary quantity/secondary quantity/tertiary quantity/base quantity

Target quantity

Target quantity of the operation

Unit

Unit of primary quantity

Yield

Yield that has been posted so far to the operation

Scrap

Scrap that has been posted so far to the operation

Rework

Rework quantity that has been posted so far to the operation

Open quantity

Open quantity that has been posted so far to the operation

Yield/target quantity [%]

Proportion of yield to target quantity in %

Yield since logon

Yield since the operation is logged on

"Staff logged on" detail application

The detail application “staff logged on” shows all persons who are logged on to the workplace selected in the detail application “workplace”. The following list describes the data available in the table. If this data is not shown by default it may be added using the column selection function.

Workplace

Workplace

Workplace to which the operation is logged on.

Person

Name

The person’s name as defined in the HR master.

First name

The person’s first name as defined in the HR master.

Name

The person's complete name as defined in the HR master (last name, middle name and first name)

Company

Company the person is assigned to in the HR master.

Personnel number

Unique key to identify the person. (Key)

Staff badge no.

Staff badge number assigned to this person in the HR master.

Operator position

Abbreviation of the operator position to which this person is logged on to at the machine.

Operator position

Unique key of the operator position at this machine to which this person is logged on to.

Order

Order

Order number of the operation.

Sequence

Sequence number of the OP (provided that sequences are used).

OP

Operation number

Split

Split number of the operation (provided that the split function is used).

SOP

Sub-operation umber (reserved)

OP designation

Designation of the operation

Article

Article number produced by the operation; taken over from operation data

Logon

Date

Date when the operation was last logged on to this workplace

Time

Time when the operation was last logged on to this workplace

Advance logon flag

Flag that the person is logged on automatically when shifts change the next time.

"Resources logged on" detail application

The detail application “resources logged on” shows all resources that are logged on to the workplace selected in the detail application “workplace”. The following list describes the data available in the table. If this data is not displayed by default it can be added using the column selection function.

Workplace

Workplace

Workplace to which the operation is logged on.

Resource

Resource type

Resource type to which the resource is assigned.

Resource

Resource ID that is entered in the resource master data.

Designation

Resource designation recorded within master data.

Resource family

Resource family (internal ID) to which the resource is assigned.

Logon

Date

Date when the resource was last logged on to this workplace.

Time

Time when the resource was last logged on to this workplace.

"Maintenances" detail application

The detail application “maintenances” shows all active maintenances for the workplace that is currently selected in the selection panel. The following list shows the data available in the table. If this data is not displayed by default it can be added using the column selection function.

Maintenance

Active

Active	light green

Status

Status of maintenance activity

Green	
Blue	"blue" threshold has been exceeded
Yellow	"yellow" threshold has been exceeded
Red	"red" threshold has been exceeded

Maintenance

Maintenance name

Type

Maintenance type defined for the maintenance:

T (cycle-based),

B (operating hours),

Z (time-based)

Class

Maintenance class

Non-recurring maintenance

Flag indicating that this maintenance is only performed once.

Valid from

Start of maintenance validity. A maintenance can only fall due within the validity period.

Valid until

End of maintenance validity.

Maintenance order

Maintenance order assigned to this maintenance.

Date

Date when this maintenance was last carried out at the selected machine.

Time

Time when this maintenance was last carried out at the selected machine.

Editor

Person (user) who reset the last maintenance.

Actual cycles

Number of cycles accrued so far.

Next maintenance after

Counter reading of cycles when the next maintenance is to be performed.

Interval

Interval within which the maintenance is to be performed; from the maintenance configuration.

Actual duration

Operating hours, which have been posted so far onto the resource – according to resource type settings.

Next maintenance after

Meter reading of the operating hours counter triggering the next maintenance to become due.

Interval

Interval in hours within which the maintenance is to be performed; from the maintenance configuration.

Next maintenance on

Date when the next maintenance falls due.

Interval

Interval within which the maintenance is to be performed; from the maintenance configuration.

Info 1 - 6

Additional text 1-6 from the maintenance configuration

"Produced material" detail application

The detail application “produced material” shows output materials on the basis of batch numbers that are logged on to the workplace selected in the detail application “workplace”. The following list describes the data available in the table. If this data is not displayed by default it can be added using the column selection function.

Workplace

Workplace

Workplace to which the batch is logged on.

Material

Material

Material number of the currently produced material

Material designation

Material designation of the currently produced material; is adopted from the producing operation.

Material type

Material type of the currently producing material; is adopted from the producing OP.

Batch number

Current batch numbers from this material produced by the OP.

Quantity

Quantity

Original quantity of the batch

Remaining quantity

Remaining quantity of the batch

Quantity unit

Quantity unit in which the batch is managed.

Logon

Date

Date when the batch was last logged on to this workplace.

Time

Time when the batch was last logged on to this workplace.

Person

Person (personnel number) who changed output batches at last.

"Material in use" detail application

The detail application "material in use" shows input materials that are logged on to the workplace selected in the detail application "workplace". The following list describes the data available in the table. If this data is not displayed by default it can be added using the column selection function.

Workplace

Workplace

Workplace to which the batch is logged on.

Material

Material

Material number of the currently produced material

Material designation

Material designation of the currently produced material; is adopted from the producing operation.

Material type

Material type of the currently producing material; is adopted from the producing OP.

Batch number

Current batch number produced from this material by the OP.

Quantity

Original quantity of the batch

Remaining quantity

Remaining quantity of the batch

Quantity unit

Quantity unit in which the batch is managed.

Logon

Date

Date when the batch was last logged on to this workplace.

Time

Time when the batch was last logged on to this workplace.

Person

Person (personnel number) who changed output batches at last.

"Shift durations" detail application

The detail application “RPA distribution” shows RPA times of the workplace selected in the detail application “workplace” in a pie chart within the current shift. Moving the mouse pointer across the pie chart shows additional information (RPA abbreviation, duration or percent).

"Shift quantities" detail application

The detail application “shift quantities” shows the current shift quantities, i.e. yield, scrap in primary quantity unit, of the workplace selected in the detail application “workplace” in a bar chart.

"Cycle progression" detail application

The detail application “cycle progression” shows saved cycle values in a line chart in [sec/cycle]. The cycle progression of the workplace selected in the detail application “workplace” is displayed. By clicking a radio button the user can decide whether they want to display the current shift or the last x hours. However, x should be less than 8 hours for performance reasons.

The following limit values are displayed as lines: upper tolerance limit - UTL (red), lower tolerance limit - LTL (red), upper action limit - UAL (yellow), lower action limit - LAL (yellow). The limits are computed and displayed on the basis of the "process parameters" configuration.

Please note: The display depends essentially on the size of the detail application.

"Downtime hit list" detail application

The downtime hit list shows the top x of current downtimes (status is not production) of the currently selected workplace within the current shift or the last hours. They are represented in a horizontal bar chart.

Using the radio buttons, it is possible to show the statuses, which have so far occurred in the current shift, or the statuses of the last x hours. By another radio button, the user can configure the display according to downtime durations or the number of respective downtimes.

The TOP X input field allows for the number of statuses to be defined (preassignment: 5).

The color of status bars corresponds to the color defined for the status text within the HYDRA configuration. The status bar is displayed in gray, in case no color is defined for the status. The status text as well as a value (duration in hours or quantity) are displayed for each bar.

Toolbar

Entry

Log on

Operations can be logged on to the system using the "log on" function

Partial confirmation/upload

The "partial upload" function allows for partial uploads on operations to be recorded in the system.

Interrupt

Operations can be interrupted in the system using the "interrupt" function

Log off

Operations can be logged off from the system using the "log off" function

Terminate

Interrupted or prepared operations can be logged off from the system using the "terminate" function

Staff

Log person on

A person may be logged on to an operation/machine using the log person on function

Log person off

A person may be logged off from the corresponding operation/machine using the log person off function

3 Machine History

Summary

Menu	Production facility management --> Production facility analysis --> Machine history
Transaction code	wphi
Function authorization	wphi

Utilization

The machine history is a report for the production management. The application allows for tracking and tracing of events that need to be posted at workplaces within MES. In this context, posting events such as status changes, order, tool, and personnel postings, maintenance activities as well as measures recorded at a workplace are listed in chronological order within a table. Different selection criteria allow for the events and periods to be evaluated.

Selection criteria

The application provides the following selection criteria:

Workplace from... to ...

This selection criterion refers to the workplace within the machine/workplace master. Wildcards (*) can be used.

Group from ... to ...

This selection criterion refers to the group in the machine or workplace master. All workplaces/machines assigned to the selected group are displayed. It is possible to use wildcards.

Short name

This selection criterion refers to the short name of machines within the master data. All machines or workplaces matching the specified character string are displayed. It is possible to use wildcards.

Designation

This field refers to the designation of machines and workplaces within the machine master data. Only the machines matching the specified character string are displayed. Wildcards (*) can be used in this field.

Cost center

This selection criterion refers to the cost center defined in the machine or workplace master. All workplaces/machines assigned to the selected cost center are displayed. It is possible to use wildcards.

Company

This selection criterion refers to the company defined in the machine or workplace master. All workplaces/machines assigned to the selected company are displayed. It is possible to use wildcards.

Report group

This selection criterion refers to the evaluation groups. All workplaces/machines assigned to the selected evaluation group are displayed.

Responsibility area

This selection criterion refers to the responsibility area within the workplace/machine master. Please respect that only machines are displayed, for which the user is authorized by the corresponding responsibility areas.

Type

Selects the type of machine/workplace that is displayed in the evaluation/report. E (individual workplaces) and G (group workplaces) may be selected.

Model

Selects the workplace type. The following workplace models may be selected:

- P Workplace
- N Machine
- J Machining center
- L Line
- A Aggregate
- C CAQ inspection station
- R Reel-based manufacturing
- S Cutting unit

Show comments

If this checkbox is selected recorded comments are displayed additionally in the table.

Comment

When machine statuses are changed, the recorded comments may be restricted in this field. * is used as wildcard character. The restricted selection is not case sensitive.

Machine statuses > X minutes only

This parameter only refers to events of the "machine status" type. If the posted duration is greater than the entered value the machine status is output.

Events

The view of the multiple events may also be restricted. All events are displayed, in case they have not been restricted.

Designation	Acronym

Machine status	M_MST
Production lock	M_PSPERRE
Operation postings	A_ADE
Personnel postings	P_ADE
Default value changes	M_VORGABE
Measure	R_MASSNAHME
Resource posting	R_MELDUNG
Release of resource	R_FREIGABE
Resource status	R_STATUS
Resetting of maintenance	R_WART_RESET
Exceeding of maintenance	R_WART_EXCEEDED
DNC Upload	R_UPLOAD
DNC Download	R_DOWNLOAD
Transfer posting of resources	R_UMBUCHUNG

Please note: Posting of events depends on the customer's system as well on its use. Consequently, it might be the case that not all events listed here are relevant.

Date from ... until (shift/time)

The period of time of the data to be evaluated can be restricted using the date selection option.

The shift date is evaluated if shift(s) are selected. In case no shift is selected, all shifts are taken into consideration. Please note that a selection by shifts is only supported for order and machine data .This is not the case for resource data.

When the "time" option is chosen, the start date is selected. The two times respectively refer to the beginning or end of the above-mentioned period of dates.

Workplaces configured as "group workplaces" may only be evaluated if "time" is chosen as selection option. In case the "shift" option is chosen, nothing is displayed, as there is no relation to shifts for group workplaces.

"Machine history" detail applications

The machine history lists all events, such as status changes, order or personnel postings of a machine that occurred on the day to be evaluated or in a shift of this day. The following postings are shown in the reports/evaluations:

Postings based on machines/workplaces

Machine statuses recorded automatically (with direct machine connection) or assigned manually at the terminal, setting of the production lock or changing of default values relating to machines/workplaces (target cycle, partitioning) at the terminal or automatic configuration with operation postings. Provided that a personal badge number is entered with the posting, the person is displayed as well. When it comes to postings relating to machines, order-related data is not displayed.

Postings based on orders

Postings performed automatically (when shifts change) or manually (logon, logoff, interruption) at the terminal. The corresponding order is displayed additionally. If it is a manual posting, the person who did the posting is shown as well. If waiting period processing is active the displayed time of logging the order on represents the time of entry and may deviate from the point in time indicated in the order log record.

Postings based on persons

Automatic (when shifts change) or manual logon or logoff processes of persons at the terminal. In addition, the corresponding personnel number as well as the operation for which the person produces are displayed.

Postings based on resources

Postings made for the machine from the HYDRA tool and resource management module (HYDRA-WRM), e.g. exceeded maintenance or measures/comments may also be displayed.

The total duration of the concerned status / event is also displayed. The duration is always zero when a person or OP is logged on. On interruption / logging off of an operation or logging off of a person, the interval between the logging on and logging off is output.

Field description

The following list describes the data available in the table. If this data is not displayed by default it can be added using the column selection function.

Field description "workplace" category

Workplace

Workplace which the event refers to

Field description "event" category

Type

Image presentation on the type

Model

Assignment of recorded events. Possible values:

- Machine status
- Production lock
- Operation postings
- Personnel postings
- Default value changes
- Exceeding of maintenance
- Resetting of maintenance

Event

Classifies the event collected at the machine, which is listed in the table row.

Type	Event
Machine status	Machine status according to configuration Coloring corresponds to the colors defined within status text configuration.
Production lock	Production lock set manually production lock canceled manually
Operation postings	OP logged on OP interrupted OP logged off
Personnel postings	Person logged on Person logged off
Default value changes	Change of partitioning/change of target cycle
Exceeding of maintenance	Maintenance cycle exceeded
Resetting of maintenance	Maintenance reset

Date

Entry date of the event

Time

Entry time of the event

Duration

Period of time between the last event of this kind and the one currently displayed. The duration is only shown for the events "OP INTERRUPTED", "OP LOGGED OFF", "PERSON LOGGED OFF" as well as for machine statuses. In any other case, 0 is displayed. These durations are durations synchronized with the shop floor shift calendar, i.e. shift breaks are not included. Consequently, this value has not necessarily to correspond to the period of time between logon and logoff.

Field description "master data" category

Workplace

Unique ID defined within the workplace configuration

Designation

Machine designation as defined in the workplace configuration

Comment

Comment on the machine as defined in the workplace configuration

Group

Capacity group which the machine was assigned to

Cost center

Cost center as defined in the workplace configuration

Company

Company as defined in the workplace configuration.

Responsibility area

Responsibility area required to view this workplace as it is defined in the workplace configuration

Field description "order" category

Order type

Order type of the order, which the event was collected for

Order

Order number of the OP, which the event was recorded for

Sequence

Sequence number of the OP (provided that sequences are used).

OP

Operation number

Split

Split number of the operation (provided that the split function is used).

SOP

Sub-operation umber (reserved)

Article

Article number produced by the operation; adopted from the operation

Article designation

Article designation of the article

Field description "person" category

Person

The personnel number of the person who was logged on or off (for personnel postings only)

Last name

The person’s last name who was logged on or off (for personnel postings only)

First name

The person’s first name who was logged on or off (for personnel postings only)

Name

Entire name (last name, second name and first name) of the person who was logged on or off (for personnel postings only)

Field description "status" category

The status number as well as the status text designation are displayed in this category, provided that the event is a machine status. The resource status is displayed for events based on resources.

Status

Status number of the assigned status

Status text

Status text of the assigned status

Receiving storage location

Destination when entering a resource status change (RES_STATUS)

Field description "maintenance" category

Maintenance type

Type of the maintenance

T: based on cycles,

B: based on operating hours

Z: Based on time

Maintenance

Short text of the maintenance

Target cycles

For maintenance type T only: number of cycles until the maintenance is due again

Actual cycles

For maintenance type T only: number of cycles accrued since the maintenance interval has been reset. Value from machine data collection

Target operating hours

For maintenance type B only: number of operating hours until maintenance falls due again.

Actual hours of operation

For maintenance type B only: number of operating hours accrued since resetting the maintenance interval. Value from machine data collection.

Next date

For maintenance type Z only: time when the maintenance falls due the next time.

Processing mode

For maintenance events (RES_WART):
R = Reset

Z = threshold exceeded

A = Enabled/disabled

For changed resource statuses (RES_STATUS):

S = Change over status

Threshold 1 (in %)

Threshold until reaching due date

Threshold 2 (in %)

Threshold until reaching due date

Threshold 3 (in %)

Threshold until reaching due date

Active

“Active” flag of the maintenance activity at the time of the event.

Active (so far)

Only relevant for processing mode A: previous “active” status of the maintenance activity at the time when the maintenance activity was activated/deactivated

Editor

Editor who edited/set/reset the maintenance.

Date

Date of editing/resetting

Time

Time of editing/resetting

Field description "measure" category

Measure

Measure name

Designation

Designation (long text) of the measure

Reporting person

Person who created the measure

Party in charge

Person who has to do the measure

Date of solution

Date when the measure has to be completed

Priority

Priority of the measure

Done

Flag indicating that the measure has been settled/done

Done by

Person who marked the measure as being settled/done.

Field description "upload/download" category

DNC file

DNC file that has been processed

Dialog ID

Dialog ID that processed the file, e.g. DNC upload

Processing ID

DNC processing ID

DNC field 1

DNC user field 1

DNC field 2

DNC user field 2

DNC field 3

DNC user field 3

DNC field 4

DNC user field 4

Field description "comment" category

Comment

Comment on the event entered by the employee

Field description "changed partitioning" category

Partitioning

Partitioning

Cavity

Cavity number

Type of modification

Reduced partitioning or increased partitioning

Reason for changes

Number of the modification reason

Text for the modification reason

Text of the modification reason

Toolbar

 **Generate order**

Using the "generate order" function, orders may be created from work plans on the basis of the specified configuration.

4 Machine Time Profile

1.1 Summary

Menu	Production facility management --> Production facility analysis --> Machine time profile
Transaction code	mtpf
Function authorization	mtpf

Utilization

The machine time profile is the ideal tool for every planner, shift manager and production manager and is a report/evaluation of the production facility management function.

Integration

The machine time profile has been designed to represent the production and downtime performance of machines of the foreman area over a specified period of time. A clear, graphic bar chart shows which machine conditions were recorded at what point in time.

Selection criteria

The application provides the following selection criteria:

Workplace from... to ...

This selection criterion refers to the workplace in the machine or workplace master. Wildcards (*) can be used.

Group from ... to ...

This selection criterion refers to the group in the machine or workplace master. All workplaces/machines assigned to the selected group are displayed. Wildcards can be used.

Evaluation group

This selection criterion refers to the evaluation groups. All workplaces/machines assigned to the selected evaluation group are displayed.

Responsibility area

This selection criterion refers to the responsibility area within the workplace/machine master. Please respect that only machines are displayed, for which the user is authorized by the corresponding responsibility areas.

Short name

This selection criterion refers to the short name of machines within master data. All machines or workplaces matching the specified character string are displayed. Wildcards can be used.

Company

This selection criterion refers to the company defined in the machine or workplace master. All workplaces/machines assigned to the selected company are displayed. Wildcards can be used.

Status text

By entering a status text or a part of a status text, only those machines and workplaces are displayed that match the entered status text or the specified character string.

Status longer than x minutes

This selection criterion refers to the displayed statuses of the machines or workplaces. The graphic view only shows the statuses that were active at the machine longer than the specified period (in minutes).

Date from ... until (shift/time)

The period of time of the data to be selected can be restricted using the date selection option.

The shift date is evaluated if shift(s) are selected. In case no shift is selected all shifts are taken into consideration. Please note that a selection by shifts is only supported for order and machine data .This is not the case for resource data.

The start date is selected if the selection is made by the time. The two times respectively refer to the beginning or end of the above-mentioned period of dates.

Workplaces configured as "group workplaces" may only be evaluated provided that a selection is made via the "time". Nothing is displayed if selected by "shift" as there is not shift relation for group workplaces.

Designation

This field refers to the designation of machines and workplaces within the machine master data. Only the machines matching the specified character string are displayed. Wildcards (*) can be used in this field.

Cost center

This selection criterion refers to the cost center defined in the machine or workplace master. All workplaces/machines assigned to the selected cost center are displayed. Wildcards can be used.

RPA number (Resource Performance Account)

If one or several resource performance accounts are selected only the statuses, which were assigned to the corresponding RP accounts or the status time that was recorded on the corresponding RP accounts, are displayed in the graphic evaluation.

Display order

If this option is checked the operations that ran per machine are displayed in addition to the individual statuses within the Gantt.

If several selection criteria are used overlapping selection criteria are displayed in the workplace overview.

View criteria

In addition to selecting data within the selection criteria, the graphic representation may be changed by further view criteria:

General

In the "general" tab data may be grouped for the display. In addition to the option of defining that a grouping is to be made, a grouping option may also be indicated. The following groupings are possible at the moment:

- Group
- Cost center
- Company

Time scale

A drop-down box allows for the displayed scale to be divided into the dimensions "seconds", "minutes", "hours", "days", "weeks" and "months". The scale is displayed in the selected dimension. The checkbox "fit time scale into visible area" reduces or increases the selected time range in order for it to fit into the application (without scrolling). The "+" and "-" buttons allow for data to be increased or reduced manually or step by step.

Workplace table

This multi-select box allows for the displayed data to be selected in the left table view. The folllowing information is provided:

- Workplace
- Short name
- Rate of capacity utilization
- Reserve of rate of capacity utilization
- Cost center and
- Group

Color status

In this tab the displayed bar colors may be selected according to the RPA colors, status colors and colors for production and downtime.

Operation colors

If data is displayed according to the selection criteria, it may be shown in different colors according to the following criteria:

- Category
- Order type
- Order
- Article and
- Tool

"Machine time profile" detail applications

The machine time profile is displayed and divided into a tabular view of the selected workplaces/machines and the graphic view of status development.

Tabular overview

The table view shows the workplaces/machines including additional information, which have been selected for the graphic view. The type and grouping of data may be determined, as described in the presentation criteria.

In addition to displaying master data for the selected workplaces/machines, such as short name, cost center and group, it is also possible to display the rate of capacity utilization and the reserve for the capacity utilization rate.

Rate of capacity utilization

The rate of capacity utilization is computed from the relation between production time and total time.

Formula:

The rate of capacity utilization is always calculated on the basis of all downtimes (not only the ones displayed) compared to the total time.

Reserve for the rate of capacity utilization

The reserve for the rate of capacity utilization is computed from the proportion of displayed downtimes compared with the total time.

Gaps that are less than the typical time required per piece at a machine as well as times that cannot be avoided (e.g. "works meeting" status) cannot build a reserve for the rate of capacity utilization. Consequently, they are not taken into account if they are hidden accordingly.

Formula:

$$\frac{\sum_{n=1}^{10} BMKn - (ausgeblen\det e\ Stills\tan ds - Zeiten)}{\sum_{n=1}^{11} BMKn}$$

"Table view" context menu

Workplaces/machines

Opens the workplaces application using the tabular overview of the workplaces/machines.

Status report

Opens the status report (machine-related) application using the tabular overview of the workplaces/machines.

Graphic view

The graphic view shows which machine conditions were recorded at the individual machines at which point in time. The machine time profile has been designed to represent the production and downtime performance of machines of the foreman area over a specified period of time.

"Graphic view" context menu

Order overview

Opens the order overview application using the operations displayed in the graphic view.

Operation overview

Opens the operation overview application using the operations displayed in the graphic view.

5 Cycle Parameters

Summary

Menu	Master data → Workplaces/ machines → Cycle parameters
Transaction code	cycpa
Function authorization	mdcycl

Usage

HYDRA provides the ability to monitor cycle times within the machine data recording function without requiring HYDRA process data processing to be used.
The purpose of this function is to configure the action and tolerance limits.

Integration

Values are marked in different colors in the *Workplace overview* depending on whether the action or tolerance limit was exceeded:

Standard	Black
If the value drops below or exceeds the action limit	Blue
If the value drops below or exceeds the tolerance limit	Red

If a limit is exceeded, no further processing steps are taken in HYRDRA.

Requirement

Before defining any configurations, you must first set up the machine.

Selection criteria

The application provides the following selection criteria:

Machine

Selection by machine/ workplace

Field descriptions

Machine

Machine for which the configuration applies.

Tolerance limit positive, negative

Values may not drop below or exceed the percentage values defined here. The cycle time of the logged on operation is always used as the target value for cycle time monitoring. This can be corrected at the terminal. The limit value is entered as a percentage of the target value.

Example:	Target value:	20 sec/ cycle
	tolerance positive:	10 %
	tolerance negative:	5 %

Thus, this results in the following limit values:

Upper limit value:	22 sec/ cycle
lower limit value:	19 sec/ cycle

Action limit positive, negative

Percentage values can be entered here, triggering a warning once they have been reached. This is why the action limits should be defined more narrowly than the tolerance limits. The limit value is entered as a percentage of the target value.

6 Cycle Progression

Overview

Menu	Production facility management → Key figures → Cycle progression
Transaction code	Cycl
Function authorization	Cycl

Usage

The purpose of this overview is to show a timely presentation of a machine's cycle development over an arbitrarily selectable period of time.

Integration

The data displayed here are collected and saved as part of the machine data collection (MDE). Please also note the information about the database at the end of this section.

Requirement

Please also note the information about the database at the end of this section.

Selection criteria

The application provides the following selection criteria:

Workplace

Enter the number of the workplace/ machine, for which you would like to display a cycle progression.

Point in time ... to ...

When the application is pulled up, the point in time is predefined as follows:

From: Date = "yesterday"/ time = "now"

To: Date = "today"/ time = "now"

Choose the point in time for which the cycle progression should be displayed. Please keep in mind that the length of the point in time will affect data calculation and therefore the response time for the evaluation.

Grid

When the application is pulled up, the grid "point in time" is predefined. Choose the desired grid in which you would like the evaluation to run.

If grid spacing is chosen (not equal to "point in time"), then the calculated actual cycle is the arithmetic mean value of all random samples of actual cycles in the relevant grid spacing period:

$$\overline{IZY} = \frac{1}{n}\sum_{i=1}^{n} IZY_i$$

IZY = actual cycle

The time for the values in this case is the end time of the grid interval. Thus, for example, for hour girds, the values between 1.00 pm and 2.00 pm are calculated and 2.00 pm is displayed as the point in time.

Tabular report

Different presentation options can be chosen for the table view. The below-mentioned data is shown:

Date, time

Point in time when actual cycle data was saved. Please also note the information about the data basis at the end of this section.

Sec/ cycle (depending on what table was selected)

Shows the actual cycle in [seconds/cycle].

Cycle/ sec (depending on what table was selected)

Shows the actual cycle in [cycles/seconds].

Min/ cycle (depending on what table was selected)

Shows the actual cycle in [minutes/cycle].

Cycle/ min (depending on what table was selected)

Shows the actual cycle in [cycles/minutes].

LTL

Calculated lower tolerance limit for the selected machine based on the target cycle available when the actual cycle was saved and on the configuration cycle parameter.

Formula: LTL = Target cycle - (Target cycle * [Tolerance limit, negative] / 100)

LAL

Calculated lower action limit for the selected machine based on the target cycle available when the actual cycle was saved and on the configuration cycle parameter.

Formula: LAL = Target cycle - (Target cycle * [action limit, negative] / 100)

UAL

Calculated upper action limit for the selected machine based on the target cycle available when the actual cycle was saved and on the configuration cycle parameter.

Formula: UAL = Target cycle + (Target cycle * [action limit, positive] / 100)

UTL

Calculated upper tolerance limit for the selected machine based on the target cycle available when the actual cycle was saved and on the configuration cycle parameter.

Formula: UTL = Target cycle + (Target cycle * [Tolerance limit, positive] / 100)

The application does not show the target cycle that is active when saving an actual cycle.

Changes to data in the "machine-related postings" application do not affect this application.

Graphic detail applications

Similar to the tabular detail applications, there are four different detail applications available to show the values as a graphic display, each of which present the data in a different unit:

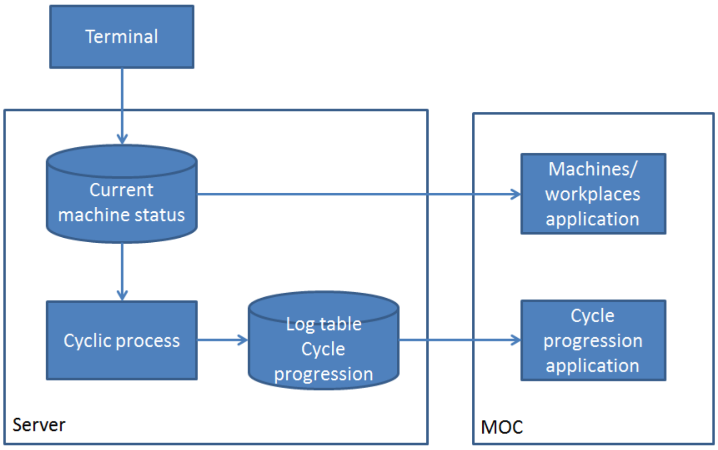
- Seconds/ cycle
- Cycles/ second
- Minutes/ cycle
- Cycles/ minute

The tolerance limits (red) and action limits (yellow) are shown in graphics.

Notes on the database for displaying the cycle progression

The current actual cycle for each of the separate machines is stored together with the current point in time and the currently set target cycle in a special log table using a cyclic process.

Schematic process:



By default, the cycle for which the process stores the data in the log table for cycle progression is set to every 30 minutes. As needed and accounting for the total capacity of the customer's system, this cycle can also be set to lower intervals (e.g. every 15 minutes).

The function cycle progression accesses values stored in the log table and displays these as a graph in the time progression.

Changes to data in the "machine-related postings" application do not affect this application.

By default, the cycle data for a machine are available for 50 calendar days. As needed and accounting for the total capacity of the customer's system (must be assured by the customer), the data for each machine can also be stored for longer (e.g. 90 days).

For both cases (modifying the logging cycle or availability duration), the respective entry must be adjusted in the Scheduler.

```
# Cycle analysis - start it every 30 minutes
L MDE-BP I    30 ./mz_zykl.out      50 > /dev/null 2> /dev/null

             ^ Logging cycle    ^ Availability period of
               (in minutes)       log data in days for
                                      cycle analyses
```

You must restart the MES after the values have been modified.